

Supplementary Materials for: ‘Familiar locations and similar activities: examining the contributions of reliable and relevant knowledge in offenders’ crime location choices’

S1. Variable construction: methodological detail

This appendix provides detailed descriptions of how the individual reliability and relevance variables, the reliability and relevance indices and combined variable, and the opportunity covariates were constructed.

Individual reliability and relevance variables

Reliability and relevance variables (e.g., recency, behavior similarity) were calculated for each prior activity node (e.g., home address, family member’s home address, location of one or more prior crimes). In developing coding rules, we distinguished between activity nodes that spanned a time period with start and end dates but no time-of-day information (offender and family home addresses, school, and work), and those that represented events that took place at a specified time and date (prior crimes, victim/witness events, non-crime incidents and the miscellaneous police contacts). Where crime, victim/witness and incident events were recorded as having occurred at an unknown time within a start and end timeframe, we used a random date-time within that timeframe to calculate similarity of timing to the reference offense. This method has been shown to predict actual offense timing most accurately in other studies (Ashby & Bowers, 2013; Boldt & Borg, 2016). Although the ‘other police contacts’ were entered with ‘start’ and ‘end’ dates in the same section of the database as the home addresses, preliminary checks showed the vast majority to relate to one-off events on the ‘start’ date, so they were treated as events.

For consistency of interpretation across the individual variables, we created them on positive scales so that the smaller the value, the lower the frequency, recency, and so on. No activity nodes could have a value of 0 for any variable because 0 would represent the absence of an activity node. We transformed variables initially calculated as differences to positive scales of proximity/similarity by taking the reciprocal. This consistency was useful for other analyses focused on the individual variables not relevant to the present paper. The reciprocal transformations were moot in the present analysis, however, given we then standardized the variables by using percentiles (as described in more detail below). For example, the same activity location would fall on the 75th percentile, whether we had taken the reciprocal, or simply reversed the percentiles to achieve a positive scale (i.e., 25th percentile of a difference becomes the 75th percentile of a similarity). However, we report the full variable construction process here for completeness.

Frequency was expressed as the proportion of days over the offender’s association with

a given node that it was visited. Offenders' home nodes were coded as 1 on the assumption that one visits home daily. Homes of their family members were coded as 0.07 representing fortnightly visits (26 visits per 365 days a year = 0.07) on the basis of research showing that most New Zealanders visit family members at least 'several times a month' (ISSP Research Group, 2009). School/education nodes were coded as 0.55 (40 weeks at 5 days a week = 200 visits per 365 days = 0.55) based on the typical school year in New Zealand and work nodes as 0.66 (48 weeks at 5 days a week = 240 visits per 365 days = 0.66) based on the typical work year. For prior crime, incident and other police contact nodes the rate was calculated as the number of unique dates on which events were recorded at the location (i.e., frequency as a count) divided by the number of days between the first and last event at the location (i.e., duration).¹

Recency represents how close in linear time the prior activities were to the reference offense. To create a variable representing temporal proximity, rather than difference, we took the reciprocal of the number of days between the reference offense and the most recent activity at that node. This created a continuous variable with maximum of 1 representing maximal recency with reference to the date of the reference offense. For example, for a home node that was current at the reference offense date (i.e., no end date or end date was after the reference offense) there was 1 day between the most recent activity and the reference offense (using 0 days would have resulted in dividing by zero), resulting in a reciprocal of $1/1=1$. A node where a prior crime had been committed a week ago was $1/7 = 0.14$. A home address recorded with an end date a year prior to the reference offense was $1/365 = 0.003$.

Duration was calculated as the length of time in days between the first recorded association with a given activity node and the latest. For former home, family home, school and work nodes, duration was the number of days between the recorded start and end dates. For such nodes with end dates after the reference offense, indicating a current association, duration was the number of days between the recorded start date and the reference offense. For offense, incident, and other police contact nodes, duration was the length of time between the first and last offense/incident/contact at that location.

Behavioral similarity between prior activities and the reference offenses was derived using the results of Kuang et al.'s (2017) study of the similarity of different crimes. Kuang et al. analyzed text descriptions of offenses in Los Angeles and calculated cosine similarity between different offense categories such as burglary, robbery, theft and assault. We used the similarity matrix presented in Kuang et al. (figure 6) to assign

¹ For most event-based nodes, only one event was recorded at the location, resulting in a rate of $1/1=1$ (maximum frequency), which is likely an overestimation of frequency for these nodes. We considered this acceptable given some overestimation was desirable to account for the fact that a single data point recorded by police likely captures only one of many visits to a location. Further, the overestimation is offset by the inclusion of the duration variable, for which these one-off event nodes score the minimum of 1 day.

equivalent NZ offense categories similarity scores in relation to burglary and robbery, which were then grouped into three categories: same offense, similar offense, and not similar offense. Similarity categories in relation to sex offenses and other offenses and police callout incidents that were not reported in Kuang et al., were assigned a priori and checked for agreement among the present authors. Victim/witness nodes were also categorized as involving the same, similar or not similar type of offense/incident.

Table S1.1 below provides detail of the resulting similarity values for all prior crime and incident types. For burglary and robbery offenses, similarity categories were assigned based on the similarity matrix of Kuang et al. (2017). For sex offenses (which were not included in their analysis), other violent and interpersonal offenses (e.g., abuse, threats and harassment) were considered similar and all other offenses/incidents were considered not similar. A spreadsheet listing the specific offense and incident codes falling within each offense/incident category is available from the corresponding author. The resulting ordinal variable was: prior offending same (7), similar (6), not similar (5), prior victimization/witness offense/incident same (4), similar (3), not similar (2), and other activity node (1).

Table S1.1 Prior offense/incident behavior similarity values relative to reference offense

Offense/ incident	Burglary	Robbery	Sex offenses
Abuse/neglect*	Similar	Not similar	Similar
Assault	Not similar	Similar	Similar
Assault (weapon)	Not similar	Similar	Similar
Breach court order	Similar	Not similar	Not similar
Breach non-association order*	Similar	Not similar	Similar
Burglary	Same	Not similar	Not similar
Disorder*	Similar	Similar	Not similar
Drugs*	Not similar	Not similar	Not similar
Embezzlement	Not similar	Not similar	Not similar
Firearms	Similar	Similar	Similar
Forgery/documents	Not similar	Not similar	Not similar
Fraud/deception	Similar	Similar	Not similar
Harassment*	Not similar	Not similar	Similar
Homicide	Not similar	Similar	Similar
Kidnap	Not similar	Similar	Similar
Mental distress*	Not similar	Not similar	Not similar
Miscellaneous*	Similar	Similar	Not similar
Property damage	Similar	Not similar	Not similar
Resist/obstruct	Not similar	Similar	Similar
Robbery	Not similar	Same	Similar
Sex	Not similar	Similar	Same
Shoplift < \$1000	Not similar	Not similar	Not similar
Shoplift > \$1000	Not similar	Not similar	Not similar
Suspicious behavior*	Not similar	Not similar	Not similar
Theft	Similar	Not similar	Not similar

Offense/ incident	Burglary	Robbery	Sex offenses
Theft from person	Not similar	Similar	Not similar
Theft from vehicle	Similar	Not similar	Not similar
Theft of bicycle	Similar	Not similar	Not similar
Theft of vehicle	Not similar	Not similar	Not similar
Threats	Not similar	Not similar	Similar
Trespass	Similar	Not similar	Not similar
Weapon	Not similar	Similar	Similar

*Prior non-crime incidents falling under these headings were assigned to the victim/witness similarity categories as it is not possible to be an offender in relation to a non-crime incident.

Location similarity reflects the type of location (rather than geographical proximity) involved in a prior activity, by comparison of the prior activity location with the location of the reference offense. Locations refer to structures and places in the built environment that have a specific function, such as schools, stores, stations, or parks, and not to whole areas like neighborhoods. Activity nodes and reference offenses were categorized into 4 location types: residential, commercial premises, public premises, and street/open spaces/transit². ‘Residential’ included private dwellings and associated outbuildings/grounds, residential construction sites, farms, rest homes. ‘Commercial’ included shops, restaurants, bars, entertainment facilities, businesses, factories, malls, offices, gyms, motels, hotels, hostels, campgrounds, commercial construction sites. ‘Public’ included school/education, sports facilities/grounds, hospitals and medical clinics, transit stations, police stations, courts, prisons, church/religious facility, community buildings, marae. ‘Street/open space/transit’ included streets, roads, footpaths, parks and open spaces, car parking facilities, wharf, harbor, beach, lake, river, sea, bus, car, train, boat, ship, plane. Comparing the location types of the activity nodes and reference offenses resulted in an ordinal variable for location similarity: same location type (3), unknown similarity—where either the reference offense or the node was missing location type information so similarity could be either same or different—(2), different location type (1).

Hour similarity represents how close in terms of hour of the day the prior activities were to the reference offense. To create a variable representing similarity, rather than difference, we took the reciprocal of the number of hours difference between the reference offense and the prior activity at each node. This created a continuous variable with maximum of 1 representing the same hour of the day ($1/1=1$, to avoid using 0 as a denominator). For example, a prior activity occurring at 13:00 and a reference offense at 01:00 are maximally different at 12 hours apart. This case would result in an hour similarity value of $1/13 = 0.08$. Home and family home nodes, where the offender could have been present at any time of day or day of week, received an hour similarity value

² Location type was identified through a range of indicators because there is no single field in NIA for it. Location type was coded as missing when no indicator enabled it to be coded as any of above (e.g., where codes such as “miscellaneous”, “other” or “online” were used).

of 1. School and work nodes received an hour similarity value of 1 if the reference offense occurred during typical school (08:00-16:00) or work (08:00-18:00) hours, otherwise a similarity value based on the difference between the start or end of the typical school/work day and the time of the reference offense. For the ‘other police contact’ nodes, which did not include time of day information, hour similarity was imputed based on the median hour similarity over all nodes.

Day similarity represents how close in terms of day of the week the prior activities were to the reference offense. Again, to create a similarity rather than difference, we took the reciprocal of the number of days of the week (from 1 indicating the same day of the week, to 4) between the prior activity and the reference offense. This created a variable with a maximum of 1 representing the same day of the week. For example, a prior activity occurring on Monday and a reference offense on Thursday are maximally different at 3 days apart. This case would result in a day similarity value of $1/4 = 0.25$. Home and family home nodes, where the offender could have been present at any time of day or day of week, received a day similarity value of 1. School and work nodes received a day similarity value of 1 when the reference offense fell on a weekday, and 0.5 when it fell on a weekend.

Season similarity was likewise the reciprocal of the number of seasons (from 1 indicating the same season, to 3 for the opposite season) between the prior activity and the reference offense (with a maximum of 1 for same season). For example, if a home address spanned more than a year, similarity = 1. If an offender only lived at a temporary home address during spring, and the offense occurred in autumn, similarity = $1/2 = 0.5$. If a prior offense occurred in summer and the reference offense occurred in winter, similarity = $1/3 = 0.33$.

If multiple events occurred at the same node, yielding multiple values for each of the relevance (similarity) variables described above, the node was assigned the maximum of these values.

Reliability and relevance indices

Prior to aggregating the above variables into reliability and relevance indices, we checked whether each variable explained independent variance in offenders’ crime location choices following a stepwise approach described in this section. The stepwise analysis followed the steps detailed in Figure S1.1, which first involved converting the individual reliability and relevance variables to categorical variables so that each SA2 could be dummy coded based on the presence (1) or absence (0) of an activity node of a given category of a given variable, within a given distance band. The categories for each variable are shown in Table S1.2. As with the main analyses, SA2s were only coded as 1 if no activity nodes were present in a shorter distance band, so that only the attributes of the nearest activity nodes are captured. Starting with a base model using the variable (all categories, all distance bands) explaining the most variance and iteratively adding whichever variable resulted in the most variance explained through its inclusion.

Iterations continued until either all variables were included in the model or adding more variables did not explain significantly more variance (as measured by a likelihood ratio test). For residential and non-residential burglary, all variables added significantly more variance. For commercial robbery, all variables except duration added significantly more variance. For personal robbery and sex offenses, all variables except day similarity added significantly more variance. Given the small number and lack of consistency of the exceptions, we elected to retain all variables in the main analysis, rather than introduce further complexity by varying the calculation of the reliability and relevance indices across crime types.

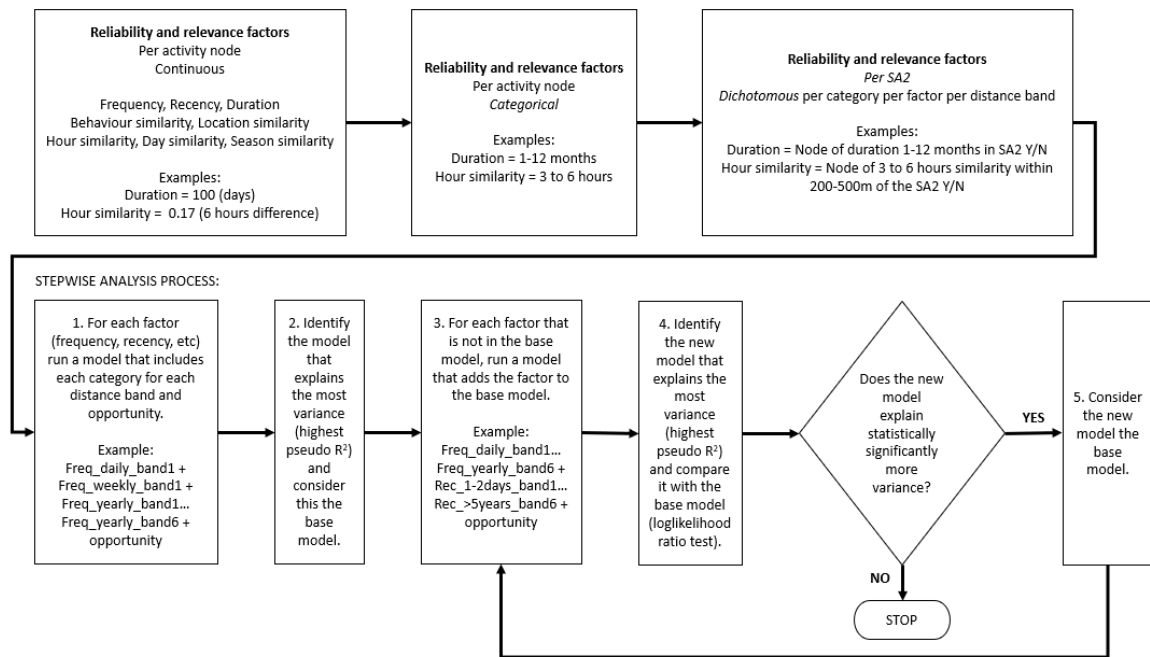


Fig. S1.1 Process for stepwise analysis of individual variables

Table S1.2 Categories for each reliability and relevance variable

Variable	Category
Frequency	Weekly or more (0.14-1) Monthly (0.03-0.14) Yearly or less (<0.03)
Recency and duration	1-2 days 3 to 30 days 1 to 12 months 1 to 5 years Over 5 years
Behavior similarity	Prior crime of same crime type Prior crime of similar crime type Prior crime of not similar crime type Prior crime experienced as a victim/witness of same crime type Prior crime or incident experienced as a victim/witness of similar type Prior crime or incident experienced as a victim/witness of not similar type

Variable	Category
Location similarity	Other activity node
	Behavior similarity unknown ³
	Same location type
	Unknown similarity
Hour similarity	Different location type
	Within 0 to 2 hours
	Within 3 to 6 hours
	Within 7 to 12 hours
Day similarity	Same day
	One day difference
	Two to three days difference
Season similarity	Same season
	One season difference
	Two seasons difference

Reliability and relevance combined variable

We reproduced the analyses using two versions of the reliability and relevance combined variable using three levels (high, medium, low), to see whether a more detailed variable (9 categories: $H_fH_s/H_fM_s \dots L_fL_s$), could reveal more detailed insights. The two versions varied the thresholds for identifying the high, medium and low categories, to test whether the results were robust to the threshold decision. The thresholds used were (1) $< 25^{\text{th}}$ percentile = low and $> 75^{\text{th}}$ percentile = high and (2) $< 33^{\text{rd}}$ percentile = low and $> 66^{\text{th}}$ percentile = high. The results (not reported) of the three-category models varied by threshold and tended to produce wide confidence intervals for some categories, likely because of the smaller numbers of nodes falling into each category by comparison to the simpler high/low version. Additionally, the three-category models produced worse fit to the data, as measured by Pseudo- R^2 . We therefore report only the results of the more robust two-category models.

Opportunity (control) variables

The opportunity covariate was different for each crime type. For residential burglary the opportunity covariate was the number of households, from New Zealand Census data. Due to large changes in residential population in many SA2s over the data period, census 2013 data was used for reference offenses occurring between 2009 and 2015, and census 2018 data for those occurring between 2016 and 2018. For the remaining crime types the opportunity variables were counts of businesses from Statistics New

³ This category was used for a small number of offender address records that indicated they were locations of prior crimes but the type of crime was not specified. There were so few of these records that the estimates had wide confidence intervals, or, in some cases, the model could not converge on any estimate. In the ordinal variable used in the main analysis this category was therefore merged with ‘other activity node’.

Zealand Business Demography data. For business counts, only the statistics for 2018 were used because they were highly correlated with those for 2013. Table S1.3 provides details of industry categories included in the counts of businesses used as opportunity variables for non-residential burglary, commercial robbery, personal robbery and sex offenses, respectively.

Table S1.3 Industry categories included in business counts for different crime types

Crime type	Industry categories included in business counts
Non-residential burglary	All industries.
Commercial robbery	G Retail Trade, H Accommodation and Food Services, K Financial and Insurance Services, L Rental, Hiring and Real Estate Services, M Professional, Scientific and Technical Services, K Financial and Insurance Services, L Rental, Hiring and Real Estate Services, M Professional, Scientific and Technical Services, N Administrative and Support Services, R Arts and Recreation Services, S Other Services
Personal robbery and sex offenses	As for Commercial robbery plus: I Transport, Postal and Warehousing, J Information Media and Telecommunications, O Public Administration and Safety, P Education and Training, Q Health Care and Social Assistance.

References

- Ashby, M. P., & Bowers, K. J. (2013). A comparison of methods for temporal analysis of aoristic crime. *Crime Science*, 2(1), 1. <https://doi.org/10.1186/2193-7680-2-1>
- Boldt, M., & Borg, A. (2016). Evaluating temporal analysis methods using residential burglary data. *ISPRS International Journal of Geo-Information*, 5(9), 148. <https://doi.org/10.3390/ijgi5090148>
- ISSP Research Group. (2009). *International social survey programme 2007: Leisure time and sports—ISSP 2007* (ZA4850 Data file Version 2.0.0). GESIS Data Archive. <https://zacat.gesis.org/webview/index.jsp?object=http://zacat.gesis.org/obj/fStudy/ZA4850>
- Kuang, D., Brantingham, P. J., & Bertozzi, A. (2017). Crime topic modeling. *Crime Science*, 6(1), 1–20. <https://doi.org/10.1186/s40163-017-0074-0>

S2. Descriptive statistics

Table S2.1 summarizes the distribution of the number of different types of activity node per offender—counting once per type of activity node per location (coordinates). For example, residential burglars had a median of 5 home addresses and 7 prior offenses.

Table S2.1 Distribution of the number of nodes per node type per offender, for each reference offence dataset

Node type	Res. Burg.			Non-res. Burg.			Com. Rob.			Pers. Rob.			Sex offenses		
	Med	IQR	Max	Med	IQR	Max	Med	IQR	Max	Med	IQR	Max	Med	IQR	Max
Home	5	9	69	5	8	73	5	8	45	5	8	69	3	5	66
Family home	18	29	352	16	27	410	20	29	300	22	35	292	9	19	282
Work	1	0	6	1	0	5	1	0	2	1	0	3	1	0	4
School	1	1	6	1	1	6	1	1	6	1	1	7	1	0	5
Offense	7	11	144	6	11	408	8	13	99	8	12	119	3	7	268
Victim/ witness	2	3	41	2	3	37	2	3	27	2	3	29	2	2	69
Incident	2	2	26	2	2	24	2	2	21	2	2	18	2	2	32
Other	6	10	152	6	11	129	7	11	96	7	12	112	4	6	223

Table S2.2 shows the percentage of all activity locations for each type of activity node and reliability/relevance category—counting once per type of activity node per location (coordinates), using the 85% of locations with a single type of activity node (excluding locations where multiple node types at a single location had been aggregated when calculating the reliability/relevance categories, for which node type and category could not be compared). To illustrate, 6.8% of residential burglars’ activity locations were H_fH_s home nodes, which was the most common category for home nodes, though the most common type of node among H_fH_s nodes was family home (18% of the total).

Table S2.2 Percentage of all activity locations per node type and reliability/relevance category, for each reference offence dataset

Offense type	Category	Home	Family home	Work	School	Offense	Victim/ witness	Incident	Other
Residential Burglary	H _f H _s	6.8	18.0	0.0	0.0	0.6	0.0	0.0	0.0
	H _f L _s	1.6	0.0	0.0	1.2	7.7	0.7	0.9	6.0
	L _f H _s	0.2	30.1	0.0	0.0	0.4	0.0	0.0	0.1
	L _f L _s	0.3	8.7	0.0	0.0	7.9	0.5	1.0	7.1
Non-residential Burglary	H _f H _s	6.4	17.6	0.0	0.3	2.7	0.1	0.1	0.9
	H _f L _s	1.5	0.0	0.0	1.1	6.4	0.7	0.7	5.3
	L _f H _s	0.2	29.1	0.0	0.0	2.4	0.1	0.2	0.9
	L _f L _s	0.4	8.0	0.0	0.0	7.0	0.5	0.9	6.5
Commercial Robbery	H _f H _s	5.3	17.7	0.0	0.0	2.2	0.1	0.1	0.8
	H _f L _s	1.4	0.0	0.0	1.5	7.3	0.7	0.5	6.4
	L _f H _s	0.1	30.1	0.0	0.0	1.9	0.1	0.1	0.7
	L _f L _s	0.2	8.3	0.0	0.0	7.2	0.6	0.9	5.6
Personal Robbery	H _f H _s	6.0	18.9	0.0	0.1	2.4	0.2	0.2	1.4
	H _f L _s	0.9	0.0	0.0	1.3	5.2	0.5	0.5	5.4
	L _f H _s	0.3	33.8	0.0	0.0	2.3	0.2	0.2	1.3
	L _f L _s	0.2	5.8	0.0	0.0	5.9	0.4	0.7	5.7
Sex offenses	H _f H _s	8.5	11.1	0.0	0.1	0.9	0.1	0.2	0.4
	H _f L _s	4.8	4.9	0.1	0.9	4.1	0.6	1.0	4.7
	L _f H _s	0.6	24.1	0.0	0.0	1.4	0.1	0.3	0.9
	L _f L _s	0.6	12.4	0.0	0.1	6.4	0.6	1.6	8.6

Table S2.3 provides descriptive statistics for the node category x distance band variables as a proportion of all SA2 alternatives sampled for each crime type. To illustrate, for the residential burglaries, 4.4% of all potentially chosen SA2 units contained a H_fH_s activity node, 3% contained a L_fL_s activity node and 20% had a H_fH_s node within 2-5km and no closer nodes.

Table S2.3 Proportion of sampled SA2s for each node category and each distance band

Category	Distance band	Res. Burg.	Non-res. Burg.	Com. Rob.	Pers. Rob.	Sex offenses
H _f H _s	Within SA2	0.044	0.046	0.041	0.046	0.030
	0-200m	0.023	0.025	0.023	0.026	0.017
	200-500m	0.031	0.033	0.032	0.035	0.024
	500-1000m	0.046	0.050	0.048	0.051	0.038
	1-2km	0.082	0.088	0.087	0.091	0.071
	2-5km	0.204	0.217	0.219	0.225	0.191
H _f L _s	Within SA2	0.023	0.020	0.022	0.018	0.022
	0-200m	0.013	0.011	0.013	0.011	0.014
	200-500m	0.015	0.013	0.016	0.013	0.018
	500-1000m	0.022	0.019	0.025	0.019	0.028
	1-2km	0.040	0.034	0.045	0.034	0.053
	2-5km	0.096	0.086	0.114	0.089	0.142
L _f H _s	Within SA2	0.038	0.040	0.037	0.044	0.025
	0-200m	0.020	0.021	0.021	0.024	0.014
	200-500m	0.027	0.029	0.029	0.033	0.019
	500-1000m	0.041	0.043	0.043	0.048	0.029
	1-2km	0.072	0.077	0.076	0.084	0.051
	2-5km	0.175	0.186	0.186	0.201	0.125
L _f L _s	Within SA2	0.030	0.027	0.025	0.022	0.026
	0-200m	0.016	0.015	0.014	0.012	0.015
	200-500m	0.020	0.018	0.017	0.015	0.020
	500-1000m	0.029	0.026	0.025	0.022	0.031
	1-2km	0.051	0.046	0.046	0.039	0.056
	2-5km	0.120	0.113	0.115	0.097	0.139

Table S2.4 describes the distribution of the opportunity variable (i.e., number of dwellings or business units) for the sampled SA2s, for each crime type. The minimum for all crime types was 0 dwellings/business units.

Table S2.4 Median, IQR and maximum number of opportunity units per SA2

Offense type	Median	IQR	Max
Residential Burglary	1167	273	4707
Non-residential Burglary	306	123	4059
Commercial Robbery	195	90	3354
Personal Robbery	231	102	3822
Sex offenses	231	102	3822

S3. Odds ratios for activity node variables

Table S3.1 shows the odds ratios (ORs) and the 95% confidence intervals for the associations between activity node proximity and crime location choice, for each activity node category, distance band and crime type. Each column represents a separate model.

Table S3.1 Odds ratios (95% confidence intervals) for each activity node category, distance band and crime

Category	Distance band	Res. Burg.	Non-res. Burg.	Com. Rob.	Pers. Rob.	Sex offenses
H _i H _s	In the SA2	117.43*** (111.75-123.41)	113.78*** (106.51-121.55)	31.37*** (26.07-37.74)	72.96*** (65.82-80.88)	321.78*** (292.78-353.65)
	0-200m	19.00*** (16.88-21.39)	33.04*** (29.10-37.52)	21.34*** (16.25-28.02)	21.69*** (17.94-26.22)	20.17*** (15.44-26.36)
	200-500m	15.13*** (13.48-16.99)	19.46*** (16.9-22.41)	12.14*** (9.02-16.34)	15.56*** (12.8-18.93)	16.46*** (12.74-21.26)
	500-1000m	8.78*** (7.77-9.93)	11.33*** (9.74-13.18)	8.02*** (5.92-10.87)	8.49*** (6.84-10.52)	14.93*** (11.89-18.73)
	1-2km	5.35*** (4.74-6.04)	7.24*** (6.24-8.39)	6.04*** (4.55-8.03)	5.61*** (4.54-6.93)	7.45*** (5.9-9.41)
	2-5km	2.9*** (2.58-3.26)	3.77*** (3.26-4.35)	3.8*** (2.93-4.93)	3.18*** (2.6-3.89)	5.19*** (4.28-6.28)
H _i L _s	In the SA2	9.72*** (9.24-10.22)	10.44*** (9.75-11.18)	12.48*** (10.37-15.03)	7.6*** (6.84-8.45)	7.65*** (6.77-8.65)
	0-200m	16.02*** (14.12-18.17)	8.91*** (7.57-10.49)	9.71*** (7.05-13.36)	10.53*** (8.4-13.2)	23.75*** (18.31-30.81)
	200-500m	11.04*** (9.68-12.59)	10.29*** (8.61-12.29)	9.33*** (6.63-13.12)	8.55*** (6.69-10.93)	18.21*** (14.11-23.5)
	500-1000m	8.39*** (7.32-9.63)	5.51*** (4.51-6.74)	4.87*** (3.38-7.03)	6.02*** (4.62-7.83)	11.19*** (8.74-14.32)
	1-2km	6.8*** (5.98-7.75)	5.13*** (4.25-6.18)	2.78*** (1.93-3.99)	3.88*** (2.97-5.08)	7.6*** (5.99-9.65)

Category	Distance band	Res. Burg.	Non-res. Burg.	Com. Rob.	Pers. Rob.	Sex offenses
	2-5km	3.62*** (3.19-4.11)	3.28*** (2.75-3.92)	2.62*** (1.94-3.54)	2.61*** (2.03-3.36)	3.54*** (2.87-4.36)
L _f H _s	In the SA2	2.33*** (2.21-2.45)	2.46*** (2.29-2.64)	3.03*** (2.53-3.64)	2.96*** (2.66-3.3)	2.96*** (2.61-3.36)
	0-200m	3.28*** (2.82-3.81)	4.66*** (3.95-5.49)	3.9*** (2.73-5.55)	3.5*** (2.74-4.48)	4.77*** (3.23-7.03)
	200-500m	3.66*** (3.16-4.24)	3.5*** (2.89-4.25)	3.12*** (2.12-4.59)	3.19*** (2.46-4.13)	3.36*** (2.25-5.03)
	500-1000m	3.12*** (2.68-3.63)	2.87*** (2.35-3.5)	4.15*** (2.92-5.89)	3.17*** (2.45-4.09)	3.25*** (2.27-4.65)
	1-2km	2.06*** (1.77-2.39)	1.77*** (1.45-2.16)	2.33*** (1.65-3.28)	2.32*** (1.8-2.99)	2.77*** (1.97-3.89)
	2-5km	1.33*** (1.16-1.52)	1.24* (1.03-1.48)	1.21 (0.88-1.65)	1.2 (0.94-1.53)	1.34 (0.99-1.82)
	In the SA2	1.68*** (1.59-1.77)	1.9*** (1.76-2.04)	2.34*** (1.94-2.81)	1.94*** (1.73-2.17)	2.06*** (1.8-2.35)
	0-200m	3.37*** (2.89-3.91)	2.59*** (2.14-3.13)	2.64*** (1.78-3.91)	2.81*** (2.11-3.73)	3.97*** (2.8-5.63)
L _f L _s	200-500m	2.83*** (2.41-3.33)	1.81*** (1.42-2.31)	3.18*** (2.09-4.83)	2.75*** (2.02-3.75)	3.95*** (2.83-5.52)
	500-1000m	2.42*** (2.05-2.86)	2.88*** (2.31-3.6)	2.1*** (1.36-3.25)	2.45*** (1.79-3.36)	3.05*** (2.22-4.19)
	1-2km	1.82*** (1.55-2.14)	1.68*** (1.34-2.11)	2.29*** (1.56-3.35)	1.49* (1.07-2.08)	2.02*** (1.47-2.77)
	2-5km	1.53*** (1.33-1.77)	1.25* (1.02-1.54)	1.34 (0.95-1.9)	1.30 (0.97-1.74)	1.93*** (1.51-2.47)

*** p<.001, *p<.05.

S4. Wald test results

Table S4.1 shows the results of Wald Chi Square tests relevant to hypothesis H1 (H_fH_s vs H_fL_s , H_fH_s vs L_fH_s , H_fH_s vs L_fL_s , H_fL_s vs L_fL_s and L_fH_s vs L_fL_s) comparing Odds Ratios (ORs) with overlapping 95% confidence intervals (CIs). All variable 1 ORs are greater than variable 2 ORs.

Table S4.1 Results of Wald tests comparing ORs where CIs overlapped

Offense	Distance band	Variable 1	Variable 2	χ^2
Residential Burglary	0-200m	H_fH_s	H_fL_s	2.54
Residential Burglary	0-200m	L_fH_s	L_fL_s	0.05
Residential Burglary	200-500m	L_fH_s	L_fL_s	4.45*
Residential Burglary	500m-1km	H_fH_s	H_fL_s	0.17
Residential Burglary	500m-1km	L_fH_s	L_fL_s	4.05*
Residential Burglary	1-2km	H_fL_s	H_fH_s	4.96*
Residential Burglary	1-2km	L_fH_s	L_fL_s	1.00
Residential Burglary	2-5km	L_fH_s	L_fL_s	1.68
Non-residential Burglary	500m-1km	L_fH_s	L_fL_s	0.00
Non-residential Burglary	1-2km	L_fH_s	L_fL_s	0.09
Non-residential Burglary	2-5km	H_fH_s	H_fL_s	0.99
Non-residential Burglary	2-5km	L_fH_s	L_fL_s	0.01
Commercial Robbery	Within SA2	L_fH_s	L_fL_s	3.01
Commercial Robbery	0-200m	L_fH_s	L_fL_s	1.67
Commercial Robbery	200-500m	H_fH_s	H_fL_s	0.93
Commercial Robbery	200-500m	L_fH_s	L_fL_s	0.00
Commercial Robbery	500m-1km	H_fH_s	H_fL_s	3.15
Commercial Robbery	500m-1km	L_fH_s	L_fL_s	4.62*
Commercial Robbery	1-2km	L_fH_s	L_fL_s	0.00
Commercial Robbery	2-5km	H_fH_s	H_fL_s	2.38

Offense	Distance band	Variable 1	Variable 2	X ²
Commercial Robbery	2-5km	L _f H _s	L _f L _s	0.16
Personal Robbery	0-200m	L _f H _s	L _f L _s	1.10
Personal Robbery	200-500m	L _f H _s	L _f L _s	0.42
Personal Robbery	500m-1km	H _f H _s	H _f L _s	2.84
Personal Robbery	500m-1km	L _f H _s	L _f L _s	1.26
Personal Robbery	1-2km	H _f H _s	H _f L _s	3.26
Personal Robbery	1-2km	L _f H _s	L _f L _s	3.61
Personal Robbery	2-5km	H _f H _s	H _f L _s	1.04
Personal Robbery	2-5km	L _f H _s	L _f L _s	0.14
Sex offenses	0-200m	H _f H _s	H _f L _s	0.53
Sex offenses	0-200m	L _f H _s	L _f L _s	0.43
Sex offenses	200-500m	H _f H _s	H _f L _s	0.22
Sex offenses	200-500m	L _f H _s	L _f L _s	0.33
Sex offenses	500m-1km	H _f H _s	H _f L _s	2.11
Sex offenses	500m-1km	L _f H _s	L _f L _s	0.06
Sex offenses	1-2km	H _f H _s	H _f L _s	0.01
Sex offenses	1-2km	L _f H _s	L _f L _s	1.57
Sex offenses	2-5km	L _f H _s	L _f L _s	2.90

*p < 0.05